

North Campus Open Space bird community analysis: survey years 1-8 (2017-2025)

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Abstract (Executive summary)

Introduction

Methods

Data collection

[Fill this in based on previous documentation in NCOS annual reports and in Dryad data publication]

Data cleaning

- summarize what happens in cleaning script
- standardize species names according to XYZ (specific ebird taxonomy version)

Data analysis

Species richness

Species composition (NMDS & guild trends)

Temporal trends for specific groups

-waterfowl, breeding birds (April-July, evidence of breeding in SB BBS), raptors...

Basic comparison to ebird data over same period

Results

Data set overview

Species richness over time

Comparison to ebird data

[1] "Vermilion Flycatcher"	"Horned Grebe"
[3] "American Herring Gull"	"Snowy Plover"
[5] "Band-tailed Pigeon"	"Brewer's Sparrow"
[7] "Lesser Nighthawk"	"Palm Warbler"
[9] "American Barn Owl"	"Golden Eagle"
[11] "White-throated Sparrow"	"American White Pelican"
[13] "Common Gallinule"	"Common Goldeneye"
[15] "Lawrence's Goldfinch"	"Lesser Scaup"
[17] "Brown Pelican"	"Brandt's Cormorant"
[19] "Grasshopper Sparrow"	"Royal Tern"
[21] "Yellow-crowned Night Heron"	"Black-and-white Warbler"
[23] "Black-throated Gray Warbler"	"Western Cattle-Egret"
[25] "MacGillivray's Warbler"	"Nashville Warbler"
[27] "Prairie Falcon"	"Red Crossbill"
[29] "Eurasian Wigeon"	"Black-legged Kittiwake"
[31] "Orchard Oriole"	"Red-throated Loon"
[33] "Western Warbling Vireo"	"Least Tern"
[35] "Semipalmated Sandpiper"	"Baird's Sandpiper"
[37] "Rufous-crowned Sparrow"	"Swainson's Hawk"
[39] "White-winged Dove"	"Brant"
[41] "Lucy's Warbler"	"Hermit Warbler"
[43] "Pine Siskin"	"Olive-sided Flycatcher"
[45] "Pelagic Cormorant"	"Common Merganser"
[47] "Black Turnstone"	"Wood Duck"
[49] "American Redstart"	"Sage Thrasher"
[51] "Bobolink"	"Swamp Sparrow"
[53] "Tennessee Warbler"	"Surf Scoter"
[55] "Vesper Sparrow"	"Reddish Egret"
[57] "Stilt Sandpiper"	"Eastern Kingbird"
[59] "Lapland Longspur"	"Swinhoe's White-eye"
[61] "Iceland Gull"	"Elegant Tern"
[63] "Pacific Loon"	"Red Phalarope"
[65] "Bald Eagle"	"Common Loon"
[67] "Yellow-breasted Chat"	"Tufted Duck"

[69] "Neotropic Cormorant"

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Species composition over time

The lowest richness occurs at survey year 1 with the following years appearing more similar.

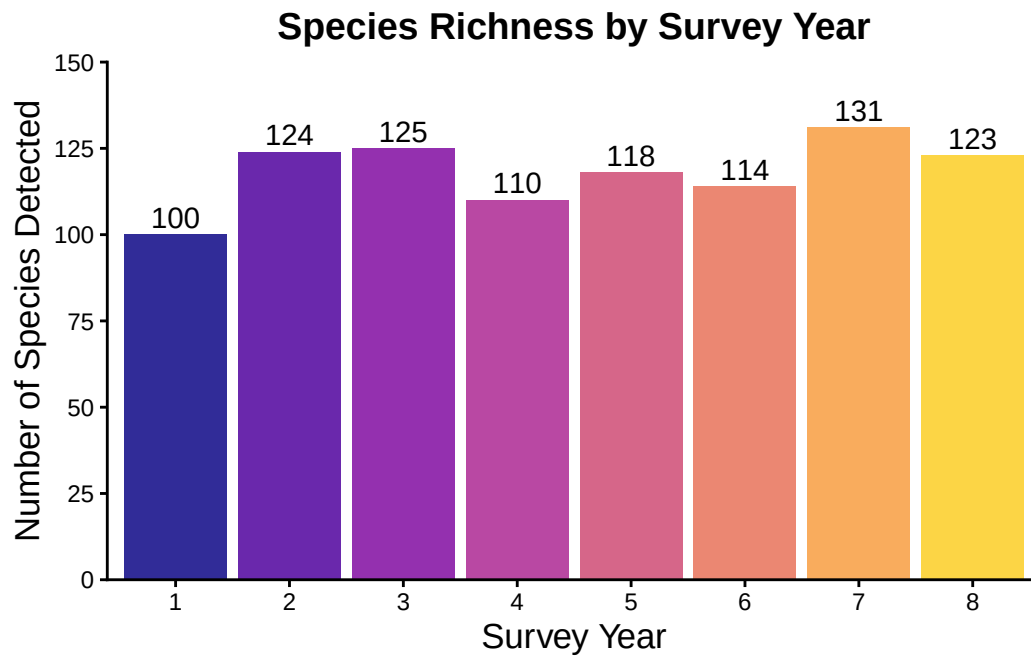
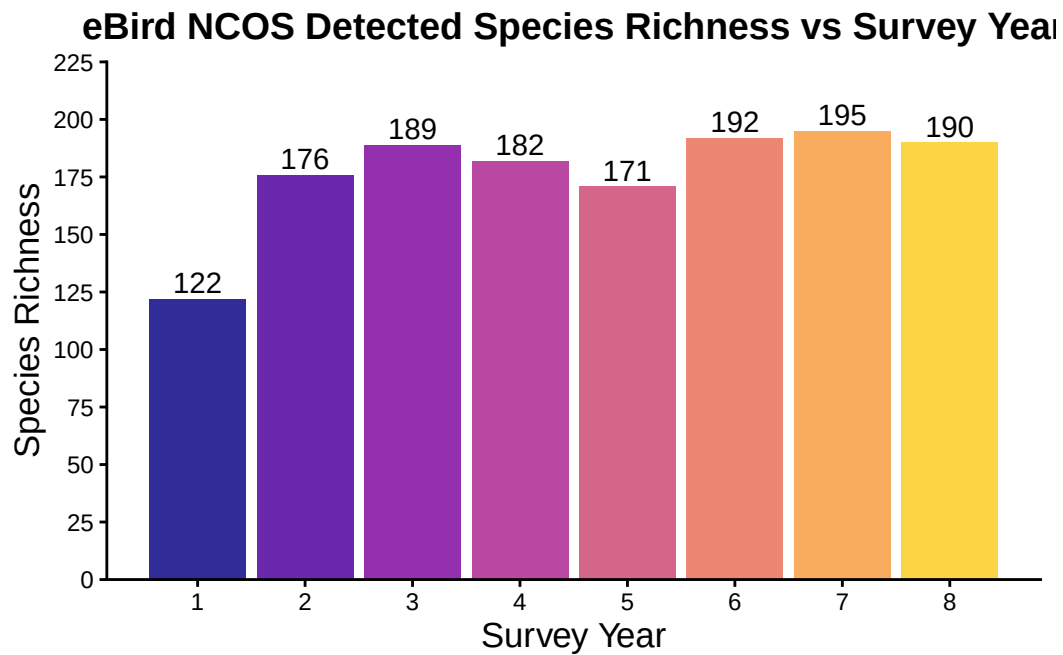
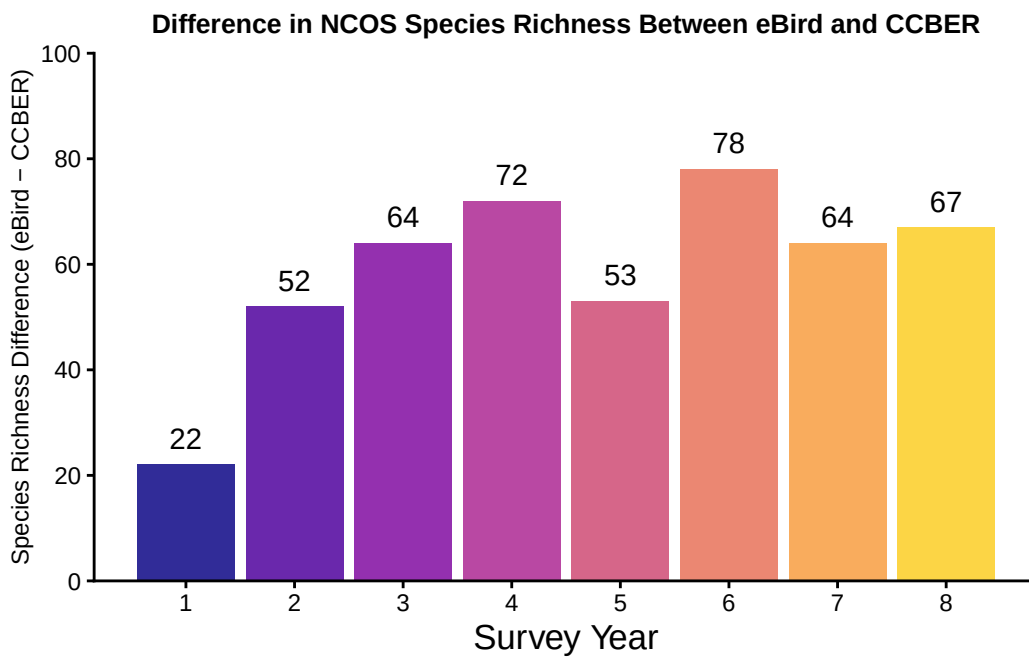
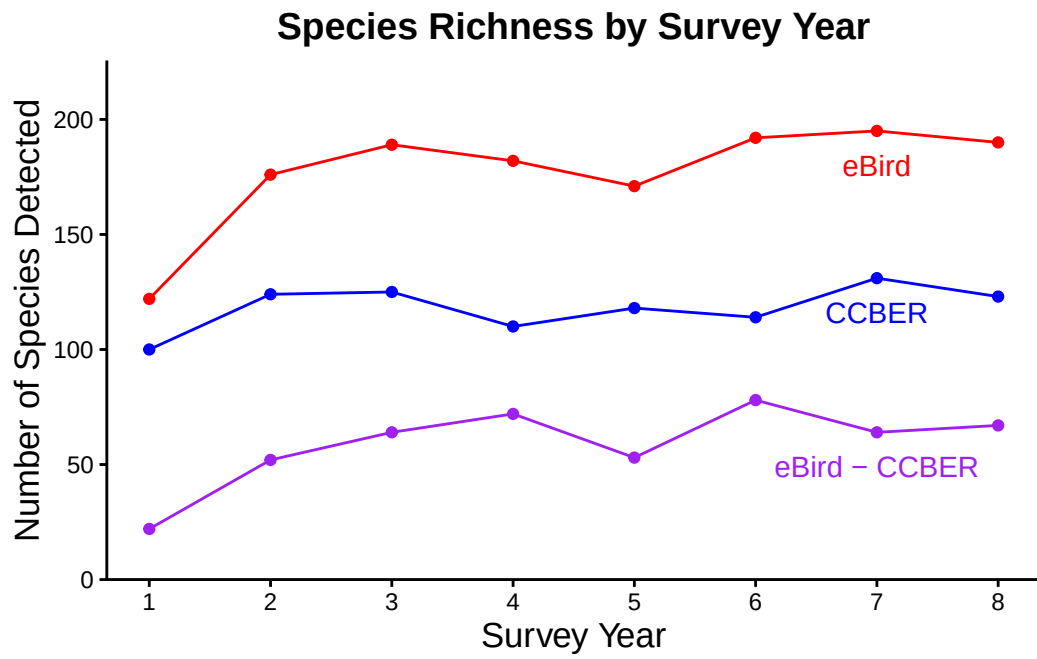


Figure 1: Species richness per survey year at NCOS. The lowest richness occurs at survey year 1 with the rest appearing more similar.



Joining with ``by = join_by(survey_year)``





Discussion